

Multiple Choice

1. **Answer:** D

Options: A) independent of M ; B) proportional to $m^{(1/2)}$; C) linear in R ; D) proportional to $R^{(3/2)}$

Note: Correct option: D - proportional to $R^{(3/2)}$

2. **Answer:** A

Options: A) 24.6 eV; B) 39.5 eV; C) 51.8 eV; D) 54.4 eV

Note: Correct option: A - 24.6 eV

3. **Answer:** B

Options: A) 2; B) 250; C) 5,000; D) 10,000

Note: Correct option: B - 250

4. **Answer:** A

Options: A) real; B) imaginary; C) degenerate; D) linear

Note: Correct option: A - real

5. **Answer:** A

Options: A) Electron; B) Meson; C) Photon; D) Boson

Note: Correct option: A - Electron

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'gamma rays'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'the electrons form a degenerate Fermi gas'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'a dye laser'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '4'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 40 m/s down. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The entropy of the system and its environment remains unchanged.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key